The Roles This Book Trains

Front Matter, Chapter 3/12 — Revision & Viva-Prep Cheatsheet (English + Roman Urdu)
Source: agentfactory.panaversity.org/docs/roles-this-book-trains

EN: Don't train syntax-writers -- train people who stand above the machine: who specify what to build, supervise the AI Workers that build it, and verify what comes back. Core pipeline: Outcome Architect -> Digital FTE Builder -> AI-Native Company Architect -> Cloud AI Engineer, field version: the vendor-neutral Forward Deployed Engineer (FDE).
Roman Urdu: Syntax-writers train mat karo -- un logon ko train karo jo machine se upar khare hain: jo specify karte hain kya banana hai, AI Workers supervise karte hain, aur wapis aaye hue ko verify karte hain. Core pipeline: Outcome Architect -> Digital FTE Builder -> AI-Native Company Architect -> Cloud AI Engineer; field version: vendor-neutral Forward Deployed Engineer (FDE).

1. The Generalist Core -- 4 Roles, One Pipeline

- **Outcome Architect** (kya) -- intent ka malik, execution ka nahi; PM ratio 1:8 se 1:20+ barha kyunke deciding bottleneck ban gayi.
- **Digital FTE Builder** (build) -- book ka primary graduate; spine: spec-driven dev, SKILL.md authoring, agent architecture, tool/MCP interfaces, evaluation, human oversight.
- **AI-Native Company Architect** (system) -- poora enterprise design karta hai (Two-Layer Model, workforce, nervous system, SoR).
- **Cloud AI Engineer** (run) -- production mein chalata hai (Azure Container Apps, Inngest, Dapr/Kubernetes).

2. The FDE -- Why Palantir Needed It

- FDE = ek customer, kai capabilities (embedded, solves whatever client needs) -- regular developer = ek capability, kai customers.
- **2x2 test:** technical product + non-technical buyer = sirf yehi cell forward deployment zaroori karta hai.
- FDE vs Solutions Architect: SA deal-close tak, FDE production tak rehta hai. Hiring test: akela hire ho jaye AND customer ke saamne bhi bhej sako.

3. Market Demand, Pay, Services Industry

- FDE postings 729-800%+ barhin (2025-26), candidate pool ~50%. Median ~$190K, senior/staff frontier labs $450-600K. MIT: ~95% custom AI pilots koi measurable return nahi dete.
- Services math: ~100 FDEs = $100M business (70-90% margins) jo traditionally 2,000-2,500 log karte the.
- **Vendor lock-in problem:** har platform FDE apne vendor ke stack mein client lock karta hai -- vendor-neutral FDE optionality bechta hai.

4. Two Systems of Record + Pod of One

- **Method SoR** (ecosystem-given, identical everywhere) + **Profession SoR** (khud-banaya, vertical/jurisdiction-specific, kisi aur ke paas nahi).
- **Pod of one:** AWS ka 5-6-engineer pod ek graduate + Digital FTEs mein simat jata hai. Risk test: 2 hafte unreachable ho to doosra repository se pick kar sake?
- **Freelance FDE:** demand supply se pehle aayi -- opening hai trained reader ke liye. Direct-hire par title AI-Native Company Architect ban jata hai, skills wahi rehti hain.

5. Skill Author, Supporting Roles, Where It Stops

- **Skill Author** = Subject Matter Expert jo tacit judgment SKILL.md mein encode karta hai -- Digital FTE ka 'knowledge engine'.
- **Connector/Plugin Engineer** = ek job, do addresses: connector-native app (claude.ai extend) vs plugin (coding agent extend).
- **3 Supporting roles:** Evals Engineer (crash-test), AI Governance Officer (limits set), Digital FTE Supervisor (day-to-day accountable).

- **Jahan book rukti hai:** LLMOps/fine-tuning (last resort, default nahi), Harness Engineer (use karte ho, banate nahi), AI Data Engineer (adjacent, central nahi).

Key Terms Cheatsheet

One line each — English + Roman Urdu.

Term	English	Roman Urdu
Outcome Architect	Owns intent and what 'correct' means -- not execution.	Intent ka malik, 'correct' ka matlab define karta hai -- execution nahi.
Digital FTE Builder	Book's primary graduate: specs, skills, tools, evals, oversight.	Book ka primary graduate -- specs/skills/tools/evals/oversight train hota hai.
AI-Native Company Architect	Designs the whole enterprise AI Workers staff.	Poora enterprise design karta hai jisme AI Workers kaam karte hain.
Cloud AI Engineer	Deploys and scales Workers in production.	Workers ko production mein deploy/scale karta hai.
Forward Deployed Engineer (FDE)	Vendor-neutral field role: embeds on-site, builds end-to-end.	Vendor-neutral field role -- on-site embed ho kar end-to-end banata hai.
Pod of One	One engineer + Digital FTEs replacing a 5-6-person team.	Ek engineer + Digital FTEs, 5-6-log ki team ki jagah.
Skill Author	A domain expert encoding tacit judgment into SKILL.md.	Domain expert jo tacit judgment SKILL.md mein encode karta hai.
Evals Engineer	Crash-tests a Worker before it goes live.	Worker ko live hone se pehle crash-test karta hai.
AI Governance Officer	Sets limits on what an AI Worker may never do.	Decide karta hai AI Worker kya kabhi na kare.
Digital FTE Supervisor	The accountable human the audit trail points to.	Accountable insaan jis tak audit trail pohanchta hai.

Viva Prep -- 10 of 66 Representative Scenario Q&A;

#	Scenario Gist	Answer	One-line Reasoning
Q1	What should be taught when nobody can name the jobs of 2036?	**Train people who specify, supervise, and verify**	Syntax fluency is the breaking old answer -- AI is absorbing syntax; judgment/spec/deployment can't be automated.
Q8	Correct order of the core pipeline?	**Outcome Architect -> Digital FTE Builder -> AI-Native Company Architect -> Cloud AI Engineer**	What -> build -> system -> run; any other order puts execution before intent.
Q15	Palantir's FDE pattern vs. a regular developer?	**One customer, many capabilities**	Regular dev = one capability, many customers; FDE embeds with one client solving whatever they need.
Q19	Why do AI vendors suddenly need thousands of embedded engineers?	**~95% of custom enterprise AI pilots show no measurable return**	MIT Project NANDA finding -- fitting AI into messy real systems is hard, not that AI doesn't work.
Q31	What do Palantir, OpenAI, Salesforce, Cursor FDEs have in common?	**Each builds on the employer's own platform, wiring it into the client**	That's the vendor lock-in pattern a vendor-neutral FDE avoids.
Q35	What two Systems of Record does a vendor-neutral FDE carry?	**The method (ecosystem-given) and the profession (self-built)**	Method is identical everywhere; profession is vertical/jurisdiction-specific and not given by anyone.
Q42	What happens to the FDE title on direct hire?	**It disappears, work continues as internal core roles**	Only the address changes (client's company becomes their own); they become AI-Native Company Architect.
Q49	Who designs the tests a Worker must pass before going live?	**The Evals Engineer**	Compared to crash-testing a car or clinical trials -- built into every chapter, not bolted on.
Q52	Should a team fine-tune a model as their default first move?	**No -- fine-tune last**	Binds you to one model snapshot and costs optionality; reach for it only when prompting/context/tools/retrieval fall short.
Q60	Does a complicated dev platform sold to engineer-CTOs need FDEs?	**No -- serve with docs and developer relations**	Bai's 2x2: forward deployment is needed only when something complicated sells to a buyer who can't implement it.

Where This Leads / Sources

Leads to: Spec-Driven Development, How to Get Paid as a Vertical FDE, How to Sell, Ecosystem chapters (FDE AF Model, System of Record). Full 66-Q quiz in 08-test-your-understanding.md.

Source: docs/roles-this-book-trains/ (this repo), content originally via Zia Tutor AI connector.